

K916 — Grace WINS the target-innocence crux (ratified), and the exact-20 splits off as one SOURCED, computable input. TARGET-INNOCENT WIN: the down quark (confined ground rung) IS the vacuum $(\lambda_1, \lambda_2) = (0, 0)$ = the self-dual center, by construction (ground = vacuum = duality fixed point, BST_CFunction_RatioTheorem 6.1, $P=1/16$). NOT placed there to yield 20 — forced by “ground = vacuum = self-dual point.” This is the independent reason the #10-theory pin needed; Casey’s writing-head intuition is now rigorous at the anchor. The exact-20 (the RATIO) is a separate, quantitative gate: only the ground rung is at the center; strange/bottom climb off by integer spectral steps, and D_3 is flat only to 2nd order ($D_3(x) \approx 3(1 - 5.83x^2)$) → the strange rung must land within $\Delta x < 0.03$ to hold 20 at 0.5%. The last input — the generation $\lambda \rightarrow x$ scale — is SOURCED/computable, not a free number: the c-function is explicit, $c_5(\lambda)/c_3(\lambda) = 1/[(2i\lambda_1 + 1/2)(2i\lambda_2 + 1/2)]$, poles at $\lambda = i/4$ ON the critical line, ratio preserves the critical line. So Elie evaluates D_3 at the actual rung positions from the explicit c-function → exact 20 (strange in the window) or BST-structured deviation.

Keeper | 2026-07-25 Sat | Grace won the hard half — the placement is target-innocent (down=vacuum=self-dual center). The exact-20 is one sourced, computable input (the $\lambda \rightarrow x$ scale). A real advance, honestly split.

The target-innocence WIN (ratified — the crux resolved for the placement)

- Grace’s #10 sourcing: the self-dual center is the VACUUM $(\lambda_1, \lambda_2) = (0, 0)$ (BST_CFunction_RatioTheorem 6.1; vacuum $P=1/16=1/2^4$).
- The confined ground rung IS the vacuum — the lowest confined state is the vacuum, which is the duality fixed point. So the down quark sits at the self-dual center BY CONSTRUCTION (“ground = vacuum = self-dual point”), NOT placed there to yield 20.
- This RESOLVES Grace’s target-innocence crux (K915) for the down-quark PLACEMENT. The independent reason the pin needed. Casey’s writing-head intuition (K914 — “the down quark is where the next commitment gets written, at least the start”) is now RIGOROUS at the anchor: writing starts at the vacuum, and the vacuum is the self-dual center.
- Ratified. The DOWN-QUARK PLACEMENT at the self-dual center is TARGET-INNOCENT.

The exact-20 splits off — a separate quantitative gate (honest constraint)

- Only the GROUND rung (down) is exactly at the center. The EXCITED rungs (strange, bottom) climb off by integer spectral steps.
- D_3 is flat only to 2nd order: $D_3(x) \approx 3(1 - 5.83x^2)$. So the flat window is NARROW — to hold $m_s/m_d=20$ at the observed 0.5%, the STRANGE rung must be within $\Delta x < 0.03$ of the center in the D_3 argument.
- ☐ the exact-20 is NOT automatic from “rungs at the self-dual point.” It rides on one number: the generation $\lambda \rightarrow x$ scale — does one spectral step (down→strange, $\Delta\lambda_1=1$) map to $\Delta x < 0.03$ (flat window → exact 20) or Δx order-1 (toward the D_3 zero at $\pi/6 \rightarrow 20$ spoiled, BST-structured deviation)?

The last input is SOURCED / computable (not a free number) — the c-function is explicit

BST_CFunction_RatioTheorem gives the explicit ratio:

$$\frac{c_5(\lambda)}{c_3(\lambda)} = \frac{1}{(2i\lambda_1 + \frac{1}{2})(2i\lambda_2 + \frac{1}{2})}$$

- Poles at $\lambda_j = i/4$ — purely imaginary, ON THE CRITICAL LINE; the ratio PRESERVES the critical line (Plancherel transport $Q^3 \rightarrow Q^5$). - So the $\lambda \rightarrow x$ scale / the D_3 factor at each rung is COMPUTABLE from the explicit c-function — a sourced closed form, not a mystery. Elie evaluates D_3 at the actual generation rung positions (from the explicit c-function) $\rightarrow$ does the strange step stay in the flat window (exact 20) or drift? - The critical-line poles reinforce the RH connection at the level of the shared c-function object (Casey's link holds structurally — NOT “proves RH”).

The down-quark tier now (honestly split)

- PLACEMENT (down at the self-dual center): TARGET-INNOCENT — down = vacuum = self-dual fixed point, sourced. WON.
- RATIO $m_s/m_d = 20$: CANDIDATE, gated on the generation $\lambda \rightarrow x$ scale (does the strange rung stay in the flat D_3 window?). One SOURCED, computable input — Elie evaluates from the explicit c-function; Grace sources the exact rung positions.
- b-quark drift (6%, Elie's flag): consistent with climbing further off the center (Casey's writing-head signature — exactness sharpest at the origin, degrading outward). The b-rung is a further spectral step $\rightarrow$ larger $\Delta x \rightarrow$ larger deviation.
- Multi-rung content (Grace's flag to Lyra): the #10-theory pin must cover not just the ground rung (won) but WHY the excited rungs (strange) stay near the flat window — the real multi-rung content of “sit at $c=1$.” Open (Lyra).

Directions (the down-quark endgame)

- ★ ELIE (the exact-20 computation): evaluate D_3 at the generation rung positions from the explicit c-function ($c_5/c_3 = 1/[(2i\lambda_1 + \frac{1}{2})(2i\lambda_2 + \frac{1}{2})]$); does the strange step (one spectral step off the vacuum) land within $\Delta x < 0.03$ (flat window $\rightarrow$ exact 20) or drift (BST-structured deviation)? Report the actual Δx . + the b-rung Δx (does it predict the 6%?).
- ★ GRACE (the rung positions): source the exact λ -positions of the {down, strange, bottom} rungs relative to the vacuum (0,0) — the generation spectral steps in the c-function argument. The last input, sourced not fabricated.
- ★ LYRA (the multi-rung pin): does the geometry keep the strange rung in the flat window (a reason the spectral step is compressed), or only the ground rung at the center? The multi-rung content of the target-innocent pin.
- KEEPER: placement target-innocent (ratified); exact-20 gated on the sourced $\lambda \rightarrow x$ scale; hold the ratio at candidate; the b-drift as a predicted signature; RH connection at the shared-object tier; capstone unchanged.

— Keeper K916, 2026-07-25. Grace WINS target-innocence: the down quark (confined ground rung) = the vacuum (0,0) = the self-dual center by construction (ground=vacuum, BST_CFunction_RatioTheorem 6.1, $P=1/16$) — placement TARGET-INNOCENT, Casey's writing-head intuition rigorous at the anchor. Exact-20 splits off: only the ground is at the center; strange/bottom climb off by spectral steps; D_3 flat to 2nd order ($\approx 3(1-5.83x^2)$) $\rightarrow$ strange must be within $\Delta x < 0.03$. The $\lambda \rightarrow x$ scale is SOURCED ($c_5/c_3 = 1/[(2i\lambda_1 + \frac{1}{2})(2i\lambda_2 + \frac{1}{2})]$, poles at $i/4$ on the critical line) — Elie evaluates D_3 at the rung positions $\rightarrow$ exact 20 or BST-structured deviation. Down-quark: PLACEMENT target-innocent (won), RATIO candidate (gated on the computable $\lambda \rightarrow x$ scale). b-drift = writing-head signature. See [[Keeper_K915_down_quark_target_innocence_crux]], [[BST_CFunction_RatioTheorem]], [[Keeper_K914_Casey_insight_the_down_quark_is_the_commitment_writing_origin]].